

Instructions:

- **Due March 05, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as `ENME403_S26_HW05_yourname.pdf`.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning Objective: This homework develops students' ability to assess stability using eigenvalues and poles, analyze pole-zero structure, compute time responses from transfer functions, and verify steady-state predictions using MATLAB.

Problem 1. Consider the following systems

- URB, where $m = 1$.
 - DRB, where $m = c = 1$.
 - UO, where $m = k = 1$.
 - DO, where $m = c = 1$ and $k = 2$.
- Recall the A matrix of each system that you computed in the previous homework. Compute the eigenvalues of A and comment on the system's stability.
 - Recall the transfer function from the force to the position that you computed in the previous homework. Compute the poles of the transfer function and comment on the system's stability.

Problem 2. Consider the transfer function

$$G(s) = \frac{s^4 + 9s^3 + 33s^2 + 51s + 26}{s^5 + 11.5s^4 + 55.75s^3 + 144.125s^2 + 194.5s + 105}. \quad (1)$$

Use `tf` to create a transfer function in MATLAB.

- Use `zpk` to find the zeros and the poles of $G(s)$.
- Use `pzmap` to plot the zeros and the poles of $G(s)$.
- Use partial fraction expansions to write $G(s)$ as a sum of simpler fractions. Do this by hand. Use `residue` to check your answer.
- Using the inverse Laplace formulas, write down the impulse response of $G(s)$. Plot your solution in MATLAB. Use `impz` to compute the impulse response, and plot it on the same figure.
- Using the FVT, predict the asymptotic output for a unit-step input. Use `step` to compute the step response. Does the asymptotic value computed by Matlab agree with the value predicted by FVT?